

Checking the $k = 9$, $\ell = 6$ tables for the search of exp

What `code/exptable9` checks on each line, in the notation of `doc/htr.md`

Paul Zimmermann's search for the hard-to-round cases of `exp` (`doc/htr.md`) reads tables with one line per subrange of inputs. This note is about his tables for every binary64 input, with Taylor polynomials of $k = 9$ terms and coefficients of $\ell = 6$ words. It says what the tables hold, the six conditions under which the search misses no hard case on a line, and how the Rocq files in `code/exptable9/` check them with one boolean function whose answer `true` is proved to imply the six conditions. The same method was used on the smaller table `in_lt` (`doc/exptable-note.typ`); this note is complete by itself.

1 Proof by a checker, for a reader who does not use Rocq

Rocq is a proof assistant: it accepts a theorem only with a proof it can verify step by step. It is also a programming language, and it can run its own programs. The files here use both sides:

- `check_raw` is a **program**. It reads lines of a table, as text, and answers `true` or `false`. It uses integers only, plus one function of the Interval library that returns two numbers $L \leq \exp(x) \leq U$.
- `check_rawP` is a **theorem** about that program: **if `check_raw` answers `true`, then every line satisfies the six conditions**. It is proved once, for every possible table.
- To prove that a table file satisfies the six conditions, Rocq runs the program on it, sees `true`, and applies the theorem.

So one does not need to trust the program, or to read it: what one must read is the statement of the six conditions (section 4). The run is done by the evaluator `native_compute`, which compiles the program to machine code.

Two words used below. The **binade** of a positive number y is the interval $[2^e, 2^{e+1})$ that contains it. An **enclosure** of y is a pair of numbers $L \leq y \leq U$, here with $U - L$ tiny compared with y .

2 The tables

The archive `all.tar.bz2` holds 2 203 files and 687 184 lines. The file `ine` is for the inputs x with $\exp(x)$ in $[2^{e-1}, 2^e)$: from `in1024`, where $\exp(x)$ reaches the largest double, down to `in-1074`, where $\exp(x)$ is below the smallest positive subnormal double; around $x = 0$, where the subranges are short, the binades are split into several files (`in0-0`, `in0-1`, ...). Most files have 294 lines, those near $x = 0$ have 257 to 665.

A line is x_0 (a double, in hexadecimal), then n , then $k = 9$ integers $B_0, \dots, B_8$ of $\ell = 6$ words of 64 bits each, the least significant word first: 56 fields. For instance, the first line of `in2` starts

`0x1.62e42fefa39fp-1 6782714940072 0xb2a413074401254d 0x6ac68b078499736e ...`

Over the 687 184 lines (all measured with Python on the files):

- x_0 is always a normal double, positive or negative (351 115 lines are negative), with $|x_0|$ from 2^{-53} to 709.8;
- a few lines have no fraction digits, like `-0x1p-17`, and some have $n = 0$;
- no subrange $[x_0, x_1]$ leaves the binade of x_0 .

3 The notation, and what changes from `in_lt`

- $\beta = 2^{64}$ is the word basis, ℓ the number of words of a coefficient.
- A subrange is $[x_0, x_1]$ with $x_1 = x_0 + nu$, where $u = \text{ulp}(x)$ is the same for all x in it. The search looks at the $n + 1$ inputs $x_0 + ju$, $j = 0, \dots, n$.
- $v = \frac{1}{2} \text{ulp}(\exp x)$, the same for all x in $[x_0, x_1]$.

- $a_i = \exp(x_0)u^i/i!$, $0 \leq i < k$, the Taylor coefficients.
- A_i is an integer with $0 \leq A_i < \beta^\ell$ and $|A_i - \beta^\ell \text{frac}(a_i/v)| < 1$, and $P(j) = A_0 + A_1j + \dots + A_{k-1}j^{k-1}$.
- A line holds $B_i = P(i) \bmod \beta^\ell$ for $i < k$: the values of P at $0, \dots, k-1$, not the A_i (step 3 of the search of `doc/htr.md`).
- $m = 43$: an input is hard to round when $\exp(x)/v$ is within 2^{-m} of an integer. E is the window of the search.

Three things are more general than for `in_lt`, where every x_0 was positive and in $[2^9, 2^{10})$:

- **The sign of x , and u , vary.** A line gives $x_0 = S_0 \cdot 2^{e_x-52}$ with S_0 an integer, $2^{52} \leq |S_0| < 2^{53}$, negative when x_0 is; then $u = 2^{e_x-52}$ and the inputs are $x_0 + ju = (S_0 + j)2^{e_x-52}$. Both S_0 and e_x are read from the line.
- **v for small outputs.** If $2^e \leq \exp(x) < 2^{e+1}$ with $e \geq -1022$, the doubles near $\exp(x)$ are the multiples of 2^{e-52} , and $v = 2^{e-53}$. Below 2^{-1021} , the subnormal doubles and those of the smallest normal binade are all multiples of 2^{-1074} : there $v = 2^{-1075}$ whatever e . Both cases are

$$v = 2^{\max(e, -1022) - 53}.$$

- $k = 9$, $\ell = 6$, and E (section 5).

4 The six conditions

For one line, the search misses no hard case when:

1. **u is constant:** $0 \leq n$, and S_0 and $S_0 + n$ have the same sign and absolute values in $[2^{52}, 2^{53})$, so every $x_0 + ju$ has exponent e_x ;
2. **v is constant:** $2^e \leq \exp(x_0)$ and $\exp(x_1) < 2^{\max(e, -1022)+1}$, so $v = 2^{\max(e, -1022) - 53}$ fits every $\exp(x)$ of the subrange;
3. **(H_A):** there are $A_0, \dots, A_{k-1}$ with $0 \leq A_i < \beta^\ell$ and $|A_i - \beta^\ell \text{frac}(a_i/v)| < 1$, and the line holds $B_i = P(i) \bmod \beta^\ell$ for $i < k$;
4. **(H_T):** for $0 \leq j \leq n$,

$$|\exp(x_0 + ju)/v - (a_0 + a_1j + \dots + a_{k-1}j^{k-1})/v| \leq \rho, \quad \rho = \exp(x_1) \frac{(nu)^k}{k!v};$$

5. **(H_E):** $E \geq \beta^\ell(2^{-m} + \rho) + (1 + n + \dots + n^{k-1})$;
6. **the window is less than half the modulus:** $2E < \beta^\ell$.

Conditions 3 to 6 are the hypotheses of `hscan_exp` in `code/APaul/rocq/TaylorLink.v`, which says the search then misses no $x_0 + ju$ with $\exp(x_0 + ju)/v$ within 2^{-m} of an integer. In Rocq, the six conditions are `line_ok S0 ex n B` in `ExpCheck.v` (annex A).

5 The parameters, and the value of E

They are the first definitions of `ExpCheck.v`:

```
Definition beta : Z := 2 ^ 64.      (* the machine word basis      *)
Definition l : Z := 6.              (* words of an A_i or a B_i    *)
Definition k : nat := 9.            (* terms of the Taylor polynomial *)
Definition m : Z := 43.             (* identical bits after the round bit *)
Definition E : Z := 0x600001 * 2 ^ 320. (* the window (inferred) *)
Definition guard : Z := 64.         (* extra bits in the exp enclosure *)
```

E is not in the tables, and the program that reads these tables is not in the repository: **E is inferred from the tables**. Over the 687 184 lines, the two error terms of condition 5 each reach 2^{-43} exactly and never more (computed with Python from n and e_x alone, bounding $\exp(x_1)/v < 2^{54}$):

term	largest value	line
------	---------------	------

$(1 + n + \dots + n^8)\beta^{-\ell}$	$2^{-43.0000}$	in-1, $x_0 = -0x1.62e42fefa39efp+0$
$\rho \leq 2^{54}(nu)^9/9!$	$2^{-43.0000}$	in-2, $x_0 = -0x1.0a2b23f3bab73p+1$

So the n of each line is chosen to keep both terms below 2^{-43} , and the window must hold three terms of 2^{-43} : $E \geq \beta^\ell \cdot 3 \cdot 2^{-43}$. The search program `htr3.c` for `in_lt` writes its window as a constant added to the top word, rounded up: `ERR = 0x400001` for $\lceil 2^{64}(2^{-43} + 2^{-43} + 2^{-90}) \rceil$. The same rule here gives $\lceil 2^{64} \cdot 3 \cdot 2^{-43} \rceil = 0x600000$, and `0x600001` with the same rounding up, so $E = 0x600001 \cdot 2^{320}$ (about $2^{342.585}$). Condition 5 is checked on every line with this value; a wrong guess would make the check fail, never pass. The value of `htr3.c`, `0x400001`, is too small here: the lines near $x = \pm 1$ fail condition 5 with it (while their B_i are right).

6 How one line is checked: `check_line`

`check_raw` reads a line with `parse` into (S_0, e_x, n, B) , then runs `check_line`, with integers only, except for two calls to the certified exp of the Interval library.

1. **Condition 1:** integer tests on n , $|S_0|$, $|S_0 + n|$ and the sign of $S_0(S_0 + n)$.
2. **Enclosures.** `encl` builds the double $S_0 2^{e_x - 52}$ as a float of Interval and calls its exp at $384 + 53 + 64 = 501$ bits: $L_0 \leq \exp(x_0) \leq U_0$, and the same at x_1 .
3. **Condition 2.** e is the exponent of L_0 ; the test is $U_1 < 2^{\max(e, -1022)+1}$.
4. **Condition 3.** For each i , $T = \beta^\ell a_i / v$ is $\exp(x_0)$ times a power of two, divided by $i!$. At both ends of the enclosure the checker computes the nearest integer R of T and the integer part q of T/β^ℓ , and requires them equal; then $A_i = R - q\beta^\ell$ is within $1/2$ of $\beta^\ell \text{frac}(a_i/v)$. It computes $P(0), \dots, P(8)$ modulo β^ℓ and compares them with the B_i .
5. **Condition 4** needs no test: it is the Taylor theorem (`ExpTaylor.v`).
6. **Condition 5** is one integer inequality, with $\exp(x_1)$ replaced by U_1 .
7. **Condition 6** is checked once by `check_raw`.

7 Three lines

The first line of three files: the largest inputs, inputs near 1, and subnormal outputs. The rows S_0 to the enclosure, and the last one, are the checker's own values; x_0 in decimal and the window terms are computed with Python.

	in1024	in2	in-1074
x_0	709.089566	0.693147 (ln 2)	-745.133219
S_0	6237217781087073	6243314768165360	-6554261109157969
e_x, u	9, 2^{-43}	-1, 2^{-53}	9, 2^{-43}
n	20778149366	6782714940072	20778149366
e	1023	1	-1075
v	2^{970}	2^{-52}	2^{-1075}
enclosure of $\exp(x_0)$	502 bits, width 2	502 bits, width 2	520 bits, width 271017
ρ	$2^{-44.00}$	$2^{-58.84}$	$2^{-97.00}$
$1 + \dots + n^8$	$2^{274.20}$	$2^{341.00}$	$2^{274.20}$
window needed	$2^{341.59}$	$2^{342.00}$	$2^{341.00}$
<code>check_raw</code>	true	true	true

The enclosure is $L_0 = p \cdot 2^f$ with p of the number of bits shown, and $U_0 = L_0 + w \cdot 2^f$ with w the width shown. The target term $\beta^\ell 2^{-43} = 2^{341}$ is in every window. At `in1024` the Taylor term ρ is near its bound 2^{-43} ; at `in2` the rounding of the A_i is, since n is large; at `in-1074`, $\exp(x_0)$ is in $[2^{-1075}, 2^{-1074})$, so $\exp(x)/v$ stays between 1 and 2, and ρ is tiny. All three need less than $E \approx 2^{342.585}$.

8 Line 1 of in-1074, step by step

The first line of in-1074 is (the B_i are shown by their most significant word only; each has six):

$$x_0 = -0x1.74910d52d3051p+9 \approx -745.133219, \quad n = 20778149366,$$

$B_0 = 0x1bec60 \dots$	$B_1 = 0x3bec60 \dots$	$B_2 = 0x5bec60 \dots$
$B_3 = 0x7bec60 \dots$	$B_4 = 0x9bec60 \dots$	$B_5 = 0xbbec60 \dots$
$B_6 = 0xdbec60 \dots$	$B_7 = 0xfbec60 \dots$	$B_8 = 0x11bec60 \dots$

It is the case `in_lt` does not have: x is negative and $\exp(x)$ is below the smallest positive double. The run of section 10 checks this line as it checks all the others: `check_raw` reads the text with `parse`, then gives the numbers to `check_line`. The numbers below are the checker's own values when it says so; the others are computed with `check_line.py`'s integer `exp` at 1200 bits (Python, not a proof).

Reading the line. `parse` finds the sign `-`, the fraction digits `74910d52d3051`, the exponent `+9`, then n in decimal and 54 words `0x...`, grouped by six, the least significant first, into $B_0, \dots, B_8$ (section 9 states and proves this reading). So $S_0 = -(16^{13} + 0x74910d52d3051) = -6554261109157969$ and $e_x = 9$.

Condition 1: u is constant. $u = 2^{9-52} = 2^{-43}$. The inputs are $(S_0 + j)2^{-43}$ for $j = 0, \dots, n$, from $x_0 \approx -745.133219$ to $x_1 = (S_0 + n)2^{-43} \approx -745.130857$, with $S_0 + n = -6554240331008603$. Both significands are negative with absolute values in $[2^{52}, 2^{53})$, so every input has exponent 9: the search walks towards 0 without leaving the binade $(-2^{10}, -2^9]$.

Enclosures of `exp` (checker's values). Interval's `exp` at 501 bits (β^ℓ , 384 bits, plus 53, plus guard = 64) returns, for x_0 , $L_0 = p_0 \cdot 2^{-1594}$ with p_0 an integer of 520 bits and $U_0 = L_0 + 271017 \cdot 2^{-1594}$, a relative width of about 2^{-501} ; at x_1 , $L_1 = p_1 \cdot 2^{-1594}$, p_1 of 520 bits, and $U_1 = L_1 + 266797 \cdot 2^{-1594}$.

Condition 2: v is constant. L_0 is in $[2^{-1075}, 2^{-1074})$, so $e = -1075$: $\exp(x_0)$ is below the smallest normal double 2^{-1022} , and $v = 2^{\max(-1075, -1022) - 53} = 2^{-1075}$, half the distance 2^{-1074} between two subnormal doubles. The test is $U_1 < 2^{-1022+1}$; in fact $U_1 < 2^{-1074}$, so every $\exp(x)$ of the line stays in $[2^{-1075}, 2^{-1074})$: between 0 and the smallest positive double, $\exp(x)/v$ goes from 1 to $\exp(x_1)/v \approx 1.002365$.

Condition 3: the A_i , then the B_i . For each i the checker forms $T = \beta^\ell a_i/v$ at both ends of the enclosure, finds the same integer part q of T/β^ℓ and the same nearest integer R of T , and takes $A_i = R - q\beta^\ell$:

i	a_i/v	q	A_i , top word
0	$1 + 2^{-43.20}$	1	0x1bec60
1	$1.14 \cdot 10^{-13} \approx 2^{-43}$	0	0x200000
2	$6.46 \cdot 10^{-27} \approx 2^{-87}$	0	0x0
3	$2.45 \cdot 10^{-40}$	0	0x0
4	$6.96 \cdot 10^{-54}$	0	0x0
5	$1.58 \cdot 10^{-67}$	0	0x0
6	$3.00 \cdot 10^{-81}$	0	0x0
7	$4.87 \cdot 10^{-95}$	0	0x0
8	$6.92 \cdot 10^{-109}$	0	0x0

The top words of the A_i are the checker's; the other columns are Python's. Here $a_i/v = (a_0/v)2^{-43i}/i!$, and only A_0 and A_1 reach the top word. The checker then computes $P(0), \dots, P(8)$ modulo β^ℓ and finds exactly the nine B_i of the line (checker's value `true`): on the top word, $P(i) = A_0 + iA_1 + \dots$ is `0x1bec60 + i · 0x200000`, which is the column of top words above.

Note $a_0/v = \exp(x_0)/v = 1 + 2^{-43.20}$: $\exp(x_0)$ is within 2^{-43} of $v = 2^{-1075}$, the midpoint between 0 and the smallest positive double. So x_0 itself is a candidate of the search ($j = 0$). This is expected: the file starts at the first double whose $\exp$ is at least 2^{-1075} , and one step u of x moves $\exp(x)$ by a relative 2^{-43} .

Condition 4: (H_T). The Taylor theorem gives the bound, with

$$\rho = \exp(x_1) \frac{(nu)^9}{9!v} \approx 2^{-97.00},$$

tiny, because $\exp(x_1)/v$ is only about 1 here, where it is about 2^{53} for a normal output.

Condition 5: (H_E). The three terms of the window are $\beta^\ell 2^{-43} = 2^{341}$, $\beta^\ell \rho \approx 2^{287}$ and $1 + n + \dots + n^8 \approx 2^{274.20}$: about $2^{341.0000}$ in all. The checker replaces $\exp(x_1)$ by $U_1 = q_1 \cdot 2^{-1594}$, with $q_1 = p_1 + 266797$, and tests one integer inequality (`window_ok`). With $\beta^\ell \rho \leq q_1 n^9 2^g / 9!$ where $g = -1594 + 384 - 43 \cdot 9 - (-1075) = -522$ (checker's value), and $C = 2^{341} + (1 + n + \dots + n^8)$, it tests

$$q_1 n^9 + C \cdot 9! \cdot 2^{522} \leq E \cdot 9! \cdot 2^{522},$$

the window condition multiplied by $9! \cdot 2^{522}$, with $E = 0x6000001 \cdot 2^{320}$ (checker's value `true`).

Condition 6 is on the whole table: $2E = 0xc00002 \cdot 2^{320} < \beta^\ell = 2^{384}$.

On this line `check_line` answers `true`; the line then satisfies the six conditions, by the theorem.

9 The theorems

```
Theorem check_rawP ls : check_raw ls = true ->
  (2 * E < beta_l)%Z /\
  forall s, In s ls -> exists S0 ex n B,
    parse s = Some (S0, ex, n, B) /\ line_ok S0 ex n B.
```

```
Theorem parse_sound s S0 ex n B :
  parse s = Some (S0, ex, n, B) -> line_text (to_list s) S0 ex n B.
```

`line_text` (`ExpParseSpec.v`, annex D) says in plain definitions what the text of a line means: an optional `-`, `0x1`, an optional `.` followed by 0 to 13 hexadecimal digits F of value f , p , a signed decimal exponent e_x , n in decimal, then the 54 words `0x...`; and $S_0 = \pm(16^{13} + f \cdot 16^{13-|F|})$, $B_i = w_{6i} + w_{6i+1}\beta + \dots + w_{6i+5}\beta^5$.

For each table file `ine` given to `mksample.sh`, the generated file `SampleAll.v` proves

```
Lemma lines_ok_N : forall s, In s Data_N.lines ->
  exists S0 ex n B, line_text (to_list s) S0 ex n B /\ line_ok S0 ex n B.
```

(N is the file name, with `in-1074` written `inm1074` and `in0-0` written `in0_0`): **every line of the file, read as its text says, satisfies the six conditions.** No lemma is admitted; the theorems rest on the axioms of Rocq's real numbers, functional extensionality, the excluded middle, and the primitive integers and strings.

10 The sample, and running it

`sample.tar.xz` holds 22 files of the archive, 7 136 lines, about 1% of it, chosen over the whole range: `in1024`, `in1023`, `in1000`, `in900`, `in700`, `in500`, `in300`, `in100`, `in10` (large inputs), `in2`, `in1-0`, `in0-0`, `in-1` (near $x = 0$), `in-100`, `in-500`, `in-900`, `in-1000` (negative inputs), `in-1021`, `in-1022`, `in-1023`, `in-1050`, `in-1074` (around and below the smallest normal output).

```
tar xJf sample.tar.xz
./mksample.sh in1024 in1023 ... in-1074
make -j15 sample          # one Run_N.v per file, native_compute
make sample-all          # SampleAll.v
```

`Run_N.v` depends on `ExpCheck.v` and `ExpParse.v` only, so the proofs can change without running the tables again.

The sample was run on `roquableu` (an Intel Xeon E5-2667 server, 24 threads) with `make -j15 sample`; all 22 files pass. Measured:

<code>Qed</code> of a file of 294 lines	26 to 34 s
<code>Qed</code> of <code>in2</code> , <code>in-1</code> (665 lines)	57 s, 65 s
<code>make -j15 sample</code> , wall	1 min 11 s
<code>make -j15 sample</code> , processor	12 min 10 s

That is about 0.1 s of processor time a line. At that rate the whole archive, 687 184 lines, would take about 19.5 hours of processor time, about 1.3 hours on 15 workers (scaled, not measured).

The bit-flip test (`ExpMutate.v`, `make test`) makes sure every coefficient and x_0 is really checked. On the first line of `in1024`, `in2` and `in-1074`, which pass, it changes one bit of one B_i (the lowest bit, bit 192 and the top bit 383, for each of the 9 coefficients), or moves x_0 by one ulp either way, or changes bit 0, 20 or 51 of $|S_0|$ (for a negative S_0 , of its two's complement): 32 changes a line. Rocq checks by evaluation that `check_line` answers `false` on each of them, in 13.5 s on the desktop (measured).

11 What is not checked

- The rest of the archive (the sample is about 1% of it).
- That the lines of a file, or of the archive, cover the whole range of inputs (the files of `in_lt` have such a check, `ExpCover.v` in `code/exptable`; here the subranges change binade of x within a file, and the check is not written).
- The value of E of Paul's program (section 5): E is inferred.
- The search itself: that is `TaylorLink.v`, which this check feeds.

A ExpCheck.v

```
(** * Checking a table for the hard-to-round search of exp, by reflection
```

The search is the one of [doc/htr.md] (Paul Zimmermann), whose notation we keep. The inputs are cut into subranges $[x_0, x_1]$ with $x_1 = x_0 + n u$, $[u = \text{ulp } x]$ and $[v = \text{ulp}(\exp x) / 2]$ constant on each. With $[a_i = \exp x_0 u^i / i!]$, each $[A_i]$ is an integer with $[0 \leq A_i < \beta^l]$ and $[|A_i - \beta^l \text{frac}(a_i/v)| < 1]$, and $[P(j) = A_0 + A_1 j + \dots + A_{(k-1)} j^{(k-1)}]$.

A line of the table is $[(x_0, n, B)]$ with $[B_i = P(i) \bmod \beta^l]$ for $[i < k]$. Here $[x_0]$ is any normal double, positive or negative, and $[\exp x_0]$ may be subnormal. $[x_0]$ is written $[S_0 2^{(ex - 52)}]$, $[S_0]$ an integer with $[2^{52} \leq |S_0| < 2^{53}]$; then $[u = 2^{(ex - 52)}]$ and the inputs are $[x_0 + j u = (S_0 + j) 2^{(ex - 52)}]$, $[j = 0 \dots n]$.

$[line_ok]$ states the six conditions under which the search on a line misses no hard case; $[check_line]$ tests them. This file holds the definitions only; $[ExpProof]$ proves the checker correct. *)

```
From Stdlib Require Import ZArith Reals Lia Lra List.
From Flocq Require Import Core.
From Bignums Require Import BigZ.
From Interval Require Import Xreal Basic Specific_bigint Specific_ops.
From Interval Require Import Float_full.
Import ListNotations.
```

```
(** ** Parameters
```

The table and the search are described here and nowhere else, in the notation of [doc/htr.md]. *)

```
Open Scope Z_scope.
```

```
Definition beta : Z := 2 ^ 64.      (* the machine word basis          *)
Definition l : Z := 6.              (* words of an A_i or a B_i      *)
Definition k : nat := 9.            (* terms of the Taylor polynomial *)
Definition m : Z := 43.             (* identical bits after the round bit *)
Definition E : Z := 0x600001 * 2 ^ 320. (* the window (inferred) *)
Definition guard : Z := 64.         (* extra bits in the exp enclosure *)
```

```
(** ** Derived constants and the binary64 format
```

$[\beta^l = 2^{l \text{bits}}]$; a double has $[prec]$ bits, the smallest normal exponent is $[emin]$: if $[2^e \leq y < 2^{(e+1)}]$, the doubles near $[y]$ are multiples of $[2^{(\max e \text{ emin} - prec + 1)}]$, so $[v]$ is $[2^{(\max e \text{ emin} - prec)}]$. *)

```
Module SFBI2 := SpecificFloat BigIntRadix2.
```

```
Module I := FloatIntervalFull SFBI2.
```

```
Definition beta_l : Z := beta ^ l.
Definition lbits : Z := Z.log2 beta * l.
Definition prec : Z := 53.          (* bits of a double              *)
Definition emin : Z := -1022.       (* smallest normal exponent      *)
```

```

Definition iprec : SFBI2.precision :=
  SFBI2.PtoP (Z.to_pos (lbits + prec + guard)).

(** [u = 2^uexp ex] for [x0] in [[2^ex, 2^(ex+1))]. *)
Definition uexp (ex : Z) : Z := ex - (prec - 1).

(** [v = 2^vexp e] for [exp x] in [[2^e, 2^(e+1))], or below [2^emin]. *)
Definition vexp (e : Z) : Z := Z.max e emin - prec.

(** ** The conditions **)

Open Scope R_scope.

Definition bp (e : Z) : R := bpow radix2 e.
Definition fracR (r : R) : R := r - IZR (Zfloor r).

(** [a_i = exp x0 u^i / i!]. *)
Definition a (x0 u : R) (i : nat) : R := exp x0 * u ^ i / INR (fact i).

(** [f 0 + ... + f (N-1)]. *)
Fixpoint sumR (f : nat -> R) (N : nat) : R :=
  match N with 0 => 0 | S N' => sumR f N' + f N' end.

(** [P(j) = A_0 + A_1 j + ... + A_(k-1) j^(k-1)], in integers. *)
Definition Pz (A : list Z) (j : Z) : Z :=
  fold_right Z.add 0%Z
    (map (fun i => (nth i A 0 * j ^ Z.of_nat i)%Z) (seq 0 k)).

(** The six conditions on a line [(x0, n, B)], [x0 = S0 2^(ex - 52)]. *)
Definition line_ok (S0 ex n : Z) (B : list Z) : Prop :=
  let u := bp (uexp ex) in
  let x0 := IZR S0 * u in
  let x1 := IZR (S0 + n) * u in
  exists e : Z,
  let v := bp (vexp e) in
  let rho := exp x1 * (IZR n * u) ^ k / (INR (fact k) * v) in
  (* 1. u is constant: every S0 + j, j = 0 .. n, has 53 bits *)
  (0 <= n)%Z /\ (2 ^ 52 <= Z.abs S0 < 2 ^ 53)%Z /\
  (2 ^ 52 <= Z.abs (S0 + n) < 2 ^ 53)%Z /\ (0 < S0 * (S0 + n))%Z /\
  (* 2. v is constant on [exp x0, exp x1] *)
  bp e <= exp x0 /\ exp x1 < bp (Z.max e emin + 1) /\
  (* 3. (H_A), and the table holds P(0) .. P(k-1) *)
  (exists A : list Z, length A = k /\ length B = k /\
    (forall i, (i < k)%nat ->
      (0 <= nth i A 0%Z < beta_l)%Z /\
      Rabs (IZR (nth i A 0%Z) - IZR beta_l * fracR (a x0 u i / v)) < 1) /\
    (forall i, (i < k)%nat ->
      nth i B 0%Z = (Pz A (Z.of_nat i) mod beta_l)%Z)) /\
  (* 4. (H_T), the Taylor remainder *)
  (forall j, (0 <= j <= n)%Z ->
    Rabs (exp (x0 + IZR j * u) / v -
      sumR (fun i => a x0 u i / v * IZR j ^ i) k) <= rho) /\
  (* 5. (H_E), the window *)
  IZR beta_l * (bp (- m) + rho) + sumR (fun i => IZR n ^ i) k <= IZR E.

Close Scope R_scope.

```

```

(** ** The checker *)

Open Scope bool_scope.

(** The float  $[N \ 2^{ue}]$ . *)
Definition xpt (N ue : Z) : SFBI2.type :=
  Specific_ops.Float (BigZ.of_Z N) (BigZ.of_Z ue).

(** A float as its signed significand and exponent. *)
Definition getS (f : SFBI2.type) : option (Z * Z) :=
  match SFBI2.toF f with
  | Basic.Float s p e => Some (if s then Z.neg p else Z.pos p, e)
  | _ => None
  end.

(** A positive float as its significand and exponent. *)
Definition getF (f : SFBI2.type) : option (positive * Z) :=
  match SFBI2.toF f with Basic.Float false p e => Some (p, e) | _ => None end.

(** An enclosure  $[[mL \ 2^{fL}, mU \ 2^{fU}]]$  of  $[exp \ (N \ 2^{ue})]$ . *)
Definition encl (N ue : Z) : option (Z * Z * Z * Z) :=
  if SFBI2.valid_lb (xpt N ue) && SFBI2.valid_ub (xpt N ue) &&
    match getS (xpt N ue) with
    | Some (p, e) => Z.eqb p N && Z.eqb e ue
    | None => false
    end
  then
    match I.exp iprec (Float.Ibnd (xpt N ue) (xpt N ue)) with
    | Float.Ibnd lo up =>
      if SFBI2.valid_lb lo && SFBI2.valid_ub up then
        match getF lo, getF up with
        | Some (pL, fL), Some (pU, fU) => Some (Z.pos pL, fL, Z.pos pU, fU)
        | _, _ => None
        end
      else None
    | Float.Inan => None
    end
  else None.

(**  $\lfloor p \ 2^g / fk + h / 2 \rfloor$ , in integers. *)
Definition flr (p g fk h : Z) : Z :=
  if Z.leb 0 g then (2 * p * 2 ^ g + h * fk) / (2 * fk)
  else (2 * p + h * fk * 2 ^ (- g)) / (2 * fk * 2 ^ (- g)).

(**  $[p \ 2^f < 2^g]$ , for  $[p \geq 1]$ . *)
Definition lt2 (p f g : Z) : bool :=
  if Z.leb g f then false else Z.ltb p (2 ^ (g - f)).

(**  $[A_i]$ , from the enclosure of  $[exp \ x0]$ :  $[beta^l \ a_i / v]$  is
 $[p \ 2^{(f + gi)} / i!]$ ; its nearest integer  $[R]$  and its integer part
 $[q]$  (in units of  $[beta^l]$ ) must be the same at both ends. *)
Definition coefA (mL fL mU fU ve ue : Z) (i : nat) (fk : Z) : option Z :=
  let gi := lbits + ue * Z.of_nat i - ve in
  let R := flr mL (fL + gi) fk 1 in
  let q := flr mL (fL + gi - lbits) fk 0 in

```

```

let Ai := R - q * beta_l in
if Z.eqb R (flr mU (fU + gi) fk 1) &&
   Z.eqb q (flr mU (fU + gi - lbits) fk 0) &&
   Z.leb 0 Ai && Z.ltb Ai beta_l
then Some Ai else None.

(** [A_i .. A_(i+c-1)], [fk] being [i!]. *)
Fixpoint coefs (mL fL mU fU ve ue : Z) (i : nat) (fk : Z) (c : nat)
: option (list Z) :=
match c with
| 0 => Some nil
| S c' =>
  match coefA mL fL mU fU ve ue i fk,
        coefs mL fL mU fU ve ue (S i) (fk * Z.of_nat (S i)) c' with
  | Some Ai, Some As => Some (Ai :: As)
  | _, _ => None
  end
end.

(** [1 + n + ... + n^(k-1)]. *)
Definition sumn (n : Z) : Z :=
  fold_right Z.add 0 (map (fun i => n ^ Z.of_nat i) (seq 0 k)).

(** (H_E) with [exp x1 <= mU 2^fU]:
    [beta^l (2^-m + mU 2^fU (n u)^k / (k! v)) + sumn n <= E]. *)
Definition window_ok (mU fU ve ue n : Z) : bool :=
  let X := mU * n ^ Z.of_nat k in
  let g := fU + lbits + ue * Z.of_nat k - ve in
  let C := 2 ^ (lbits - m) + sumn n in
  let kf := Z.of_nat (fact k) in
  if Z.leb 0 g then Z.leb (X * 2 ^ g + C * kf) (E * kf)
  else Z.leb (X + C * kf * 2 ^ (- g)) (E * kf * 2 ^ (- g)).

Definition check_line (S0 ex n : Z) (B : list Z) : bool :=
  let S1 := S0 + n in
  let ue := uexp ex in
  Z.leb 0 n && Z.leb (2 ^ 52) (Z.abs S0) && Z.ltb (Z.abs S0) (2 ^ 53) &&
  Z.leb (2 ^ 52) (Z.abs S1) && Z.ltb (Z.abs S1) (2 ^ 53) &&
  Z.ltb 0 (S0 * S1) && Nat.eqb (length B) k &&
  match encl S0 ue, encl S1 ue with
  | Some (mL0, fL0, mU0, fU0), Some (mL1, fL1, mU1, fU1) =>
    let e := Z.log2 mL0 + fL0 in
    let ve := vexp e in
    lt2 mU1 fU1 (Z.max e emin + 1) && window_ok mU1 fU1 ve ue n &&
    match coefs mL0 fL0 mU0 fU0 ve ue 0 1 k with
    | Some A =>
      forallb (fun i => Z.eqb (nth i B 0) (Pz A (Z.of_nat i) mod beta_l))
        (seq 0 k)
    | None => false
    end
  | _, _ => false
end.

```

B ExpParse.v

```
(** * Reading the lines of the table from their text

  A line of [in_lt] is kept verbatim as a primitive string:

    -0x1.62e42fefa39efp+0 6782714940072 0x15cedf0a6382cf46 ...

  that is [x0] in hexadecimal ([-]? [0x1], then [.F] with at most 13
  digits or nothing, then [p], a sign and the exponent in decimal), [n]
  in decimal, then [k * l] words in hexadecimal, each [B_i] being [l]
  words, the least significant first. [parse] turns such a line into
  [(S0, ex, n, B)] with [x0 = S0 2^(ex - 52)]; a line of another shape
  gives [None], and [check_raw] then answers [false]. *)

From Stdlib Require Import ZArith List PrimString PrimStringAxioms Uint63.
From ExpTable9 Require Import ExpCheck.
Import ListNotations.

Open Scope Z_scope.

(** ** Characters *)

Definition c_space : int := 32%uint63.    (* ' ' *)
Definition c_0 : int := 48%uint63.        (* '0' *)
Definition c_9 : int := 57%uint63.        (* '9' *)
Definition c_a : int := 97%uint63.        (* 'a' *)
Definition c_f : int := 102%uint63.       (* 'f' *)
Definition c_x : int := 120%uint63.       (* 'x' *)
Definition c_dot : int := 46%uint63.      (* '.' *)
Definition c_p : int := 112%uint63.       (* 'p' *)
Definition c_plus : int := 43%uint63.     (* '+' *)
Definition c_l : int := 49%uint63.        (* 'l' *)
Definition c_minus : int := 45%uint63.    (* '-' *)

(** The value of a hexadecimal digit, or [None]. *)
Definition hexd (c : int) : option Z :=
  if (c_0 <=? c)%uint63 && (c <=? c_9)%uint63 then Some (to_Z (c - c_0))
  else if (c_a <=? c)%uint63 && (c <=? c_f)%uint63
  then Some (to_Z (c - c_a) + 10)
  else None.

(** The value of a decimal digit, or [None]. *)
Definition decd (c : int) : option Z :=
  if (c_0 <=? c)%uint63 && (c <=? c_9)%uint63 then Some (to_Z (c - c_0))
  else None.

(** ** Numbers *)

(** The number written with the digits [cs] in base [b], most
    significant first; [None] if a character is not a digit. *)
Fixpoint digits (d : int -> option Z) (b acc : Z) (cs : list int)
  : option Z :=
  match cs with
  | [] => Some acc
  | c :: cs' =>
```

```

    match d c with
    | Some v => digits d b (acc * b + v) cs'
    | None => None
    end
end.

Definition hexv (cs : list int) : option Z :=
  match cs with [] => None | _ => digits hexd 16 0 cs end.

Definition decv (cs : list int) : option Z :=
  match cs with [] => None | _ => digits decd 10 0 cs end.

(** A word [0xhh...h]. *)
Definition wordv (cs : list int) : option Z :=
  match cs with
  | c0 :: cx :: cs' =>
    if (c0 =? c_0)%uint63 && (cx =? c_x)%uint63 then hexv cs' else None
  | _ => None
  end.

(** The characters before the first [p], and those after it. *)
Fixpoint split_p (cs : list int) : list int * list int :=
  match cs with
  | [] => ([], [])
  | c :: cs' =>
    if (c =? c_p)%uint63 then ([], cs')
    else let (a, b) := split_p cs' in (c :: a, b)
  end.

(** A signed exponent [+d...d] or [-d...d]. *)
Definition expv (cs : list int) : option Z :=
  match cs with
  | c :: cs' =>
    if (c =? c_plus)%uint63 then decv cs'
    else if (c =? c_minus)%uint63 then
      match decv cs' with Some e => Some (- e) | None => None end
    else None
  | [] => None
  end.

(** The fraction digits [F] of [0x1.F] and the exponent: [.F] may be
    absent, and [F] has at most 13 digits (trailing zeros may be left
    out); the significand is [16^13 + F 16^(13 - |F|)]. *)
Definition fracv (cs : list int) : option (Z * Z) :=
  let (fr, ex) := split_p cs in
  match fr with
  | [] => match expv ex with Some e => Some (16 ^ 13, e) | None => None end
  | cd :: fr' =>
    if (cd =? c_dot)%uint63 && Nat.leb (List.length fr') 13 then
      match digits hexd 16 1 fr', expv ex with
      | Some f, Some e => Some (f * 16 ^ Z.of_nat (13 - List.length fr'), e)
      | _, _ => None
      end
    else None
  end.
end.

```

```

(** [x0 = [-]0x1.Fp+-E]: its signed significand [S0] and its exponent,
    [x0 = S0 2^(E - 52)]. *)
Definition x0v (cs : list int) : option (Z * Z) :=
  let (sg, cs1) :=
    match cs with
    | c :: cs' => if (c =? c_minus)%uint63 then (true, cs') else (false, cs)
    | [] => (false, cs)
  end in
  match cs1 with
  | c0 :: cx :: c1 :: cs' =>
    if (c0 =? c_0)%uint63 && (cx =? c_x)%uint63 && (c1 =? c_1)%uint63 then
      match fracv cs' with
      | Some (M, e) => Some (if sg then - M else M, e)
      | None => None
    end
  else None
  | _ => None
end.

(** ** Lines *)

(** The fields of a line, separated by single spaces. *)
Fixpoint fields (cur : list int) (cs : list int) : list (list int) :=
  match cs with
  | [] => [rev cur]
  | c :: cs' =>
    if (c =? c_space)%uint63 then rev cur :: fields [] cs'
    else fields (c :: cur) cs'
  end.

(** [w_0 + w_1 beta + ... ], the words least significant first. *)
Definition wsum (ws : list Z) : Z := fold_right (fun w a => w + beta * a) 0 ws.

(** The [B_i], each of [l] words, from the list of all words. *)
Fixpoint group (c : nat) (ws : list Z) : list Z :=
  match c with
  | 0 => []
  | S c' => wsum (firstn (Z.to_nat l) ws) :: group c' (skipn (Z.to_nat l) ws)
  end.

(** All the words, or [None] if one is not a word. *)
Fixpoint wordsv (fs : list (list int)) : option (list Z) :=
  match fs with
  | [] => Some []
  | f :: fs' =>
    match wordv f, wordsv fs' with
    | Some w, Some ws => Some (w :: ws)
    | _, _ => None
    end
  end.

Definition parse (s : string) : option (Z * Z * Z * list Z) :=
  match fields [] (to_list s) with
  | fx :: fn :: fw =>
    if Nat.eqb (List.length fw) (k * Z.to_nat l) then
      match x0v fx, decv fn, wordsv fw with

```

```

    | Some (S0, ex), Some n, Some ws => Some (S0, ex, n, group k ws)
    | _, _, _ => None
  end
else None
| _ => None
end.

```

(** ** The check on the text *)

```

Definition check_raw (ls : list string) : bool :=
  Z.ltb (2 * E) beta_l && Z.leb m_lbits &&
  forallb (fun s => match parse s with
    | Some (S0, ex, n, B) => check_line S0 ex n B
    | None => false
  end) ls.

```

C ExpProof.v

(** * A [true] answer of the checker gives the six conditions *)

```

From Stdlib Require Import ZArith Reals Lia Lra List Bool.
From Flocq Require Import Core.
From ExpTable9 Require Import ExpCheck ExpParse ExpTaylor ExpArith ExpEncl.

```

Open Scope R_scope.

(** The parameters leave room for the $[2^{-m}]$ term of the window. *)

```

Lemma m_le_lbits : (m <= lbits)%Z.
Proof. apply Z.leb_le; vm_compute; reflexivity. Qed.

```

```

Theorem check_lineP S0 ex n B :
  check_line S0 ex n B = true -> line_ok S0 ex n B.
Proof.
  unfold check_line.
  set (ue := uexp ex).
  destruct (encl S0 ue) as [[[[mL0 fL0] mU0] fU0]] eqn:h0;
    [[rewrite !andb_false_r; discriminate]].
  destruct (encl (S0 + n) ue) as [[[[mL1 fL1] mU1] fU1]] eqn:h1;
    [[rewrite !andb_false_r; discriminate]].
  set (e := (Z.log2 mL0 + fL0)%Z).
  destruct (coefs mL0 fL0 mU0 fU0 (vexp e) ue 0 1 k) as [A] eqn:hc;
    [[rewrite !andb_false_r; discriminate]].
  intros h.
  apply andb_prop in h as [h hm].
  apply andb_prop in hm as [hm hB].
  apply andb_prop in hm as [hlt hw].
  apply andb_prop in h as [h hlen].
  apply andb_prop in h as [h hsg].
  apply andb_prop in h as [h h53'].
  apply andb_prop in h as [h h52'].
  apply andb_prop in h as [h h53].
  apply andb_prop in h as [hn h52].
  apply Z.leb_le in hn; apply Z.leb_le in h52; apply Z.ltb_lt in h53.
  apply Z.leb_le in h52'; apply Z.ltb_lt in h53'; apply Z.ltb_lt in hsg.
  apply Nat.eqb_eq in hlen.

```

```

destruct (encl_ok _ _ _ _ _ h0) as [pL0 [pU0 [lo0 up0]]].
destruct (encl_ok _ _ _ _ _ h1) as [pL1 [pU1 [lo1 up1]]].
exists e; cbv zeta; fold ue.
assert (hu : 0 < bp ue) by apply bpow_gt_0.
assert (hx1 : IZR (S0 + n) * bp ue = IZR S0 * bp ue + IZR n * bp ue)
  by (rewrite plus_IZR; ring).
repeat split.
- (* 1. u is constant *)
  exact hn.
- exact h52.
- exact h53.
- exact h52'.
- exact h53'.
- exact hsg.
- (* 2. v is constant *)
  apply Rle_trans with (IZR mL0 * bp fL0); [apply log2_le; exact pL0|].
  exact lo0.
- apply Rle_lt_trans with (IZR mU1 * bp fU1); [exact up1|].
  apply lt2_ok; [lia|exact hlt].
- (* 3. (H_A), and the table holds P(0) .. P(k-1) *)
  exists A.
  destruct (coefs_ok _ _ _ _ _ _ _ _ _ eq_refl (conj lo0 up0) hc)
    as [h1A hA].
  split; [exact h1A|]; split; [exact hlen|]; split.
  + intros i hi; destruct (hA i hi) as [hA1 hA2]; split; [exact hA1|].
    exact hA2.
  + intros i hi.
    rewrite forallb_forall in hB.
    apply Z.eqb_eq, hB, in_seq; lia.
- (* 4. (H_T) *)
  intros j hj; rewrite hx1.
  apply HT_ok; [exact hu|apply bpow_gt_0|lia].
- (* 5. (H_E) *)
  apply window_okP with (mU := mU1) (fU := fU1);
    [exact m_le_lbits|exact hn| |exact hw].
  split; [apply Rlt_le, exp_pos|exact up1|.
Qed.

```

(** The same on the text of the lines: each one reads as a line that satisfies the six conditions. *)

```

Theorem check_rawP ls : check_raw ls = true ->
  (2 * E < beta_l)%Z /\
  forall s, In s ls -> exists S0 ex n B,
    parse s = Some (S0, ex, n, B) /\ line_ok S0 ex n B.

```

Proof.

```

unfold check_raw; intros h.
apply andb_prop in h as [h hall].
apply andb_prop in h as [h2E _].
split; [apply Z.ltb_lt; exact h2E|].
intros s hin.
rewrite forallb_forall in hall.
specialize (hall _ hin).
destruct (parse s) as [[[[S0 ex] n] B]]]; [|discriminate].
exists S0, ex, n, B; split; [reflexivity|].
apply check_lineP; exact hall.
Qed.

```

D ExpParseSpec.v

```

(** * What a line of the table says, and [parse] reads exactly that

  A line of [in_lt] is the text

    [-]0x1[.F]pSX N 0xW_0 0xW_1 ... 0xW_(k l - 1)

  with single spaces between the fields (none inside [x0]): an optional
  minus sign, [0x1], an optional fraction [.F] with [F] 0 to 13
  hexadecimal digits, [p], a sign [S] ('+' or '-') and [X] decimal
  digits, then [N] decimal digits and each [W_j] hexadecimal digits.
  With [ex = +-X] (the sign [S]), [x0 = +-(1 + F / 16^|F|) 2^ex] (the
  optional minus sign), a missing fraction counting as [F] empty. With
  [S0 = +-(16^13 + F 16^(13 - |F|))] this is [x0 = S0 2^(ex - 52)].
  The line gives [n = N], and [B_i = w_(i l) + w_(i l + 1) beta + ...
  + w_(i l + l - 1) beta^(l-1)] for [i < k], [w_j] being the value of
  [W_j].

  [line_text] states this; [parse_sound] proves that whatever [parse]
  returns is what the text says. A character is its code, an [int]. *)

From Stdlib Require Import PrimString PrimStringAxioms UInt63.
From ExpTable9 Require Import ExpCheck ExpParse.
From Stdlib Require Import ZArith List Lia.
Import ListNotations.

Open Scope Z_scope.

(** ** Digits and numbers *)

(** The characters [c_0], [c_a], ... are those of [ExpParse]. *)

(** [c] is the decimal digit of value [d]: '0' .. '9'. *)
Definition decdigit (c : int) (d : Z) : Prop :=
  0 <= d <= 9 /\ to_Z c = to_Z c_0 + d.

(** [c] is the hexadecimal digit of value [d]: '0' .. '9', 'a' .. 'f'. *)
Definition hexdigit (c : int) (d : Z) : Prop :=
  decdigit c d /\ (10 <= d <= 15 /\ to_Z c = to_Z c_a + (d - 10)).

(** [d_0 b^(len-1) + d_1 b^(len-2) + ... + d_(len-1)], most significant
  first. *)
Fixpoint value (b : Z) (ds : list Z) : Z :=
  match ds with
  | [] => 0
  | d :: ds' => d * b ^ Z.of_nat (length ds') + value b ds'
  end.

(** [cs] is a nonempty decimal numeral of value [v]. *)
Definition dec_text (cs : list int) (v : Z) : Prop :=
  cs <> [] /\ exists ds, Forall2 decdigit cs ds /\ v = value 10 ds.

(** [cs] is a possibly empty list of hexadecimal digits of value [v]. *)
Definition hexs_text (cs : list int) (v : Z) : Prop :=
  exists ds, Forall2 hexdigit cs ds /\ v = value 16 ds.

```

```

(** [cs] is a nonempty hexadecimal numeral of value [v]. *)
Definition hex_text (cs : list int) (v : Z) : Prop :=
  cs <> [] /\ hexs_text cs v.

(** [W] is a word [0xhh...h] of value [w]. *)
Definition word_text (W : list int) (w : Z) : Prop :=
  exists H, W = c_0 :: c_x :: H /\ hex_text H w.

(** [Xs] is a signed exponent [+X] or [-X] of value [e]. *)
Definition exp_text (Xs : list int) (e : Z) : Prop :=
  exists X v, dec_text X v /\
    (Xs = c_plus :: X /\ e = v \/ Xs = c_minus :: X /\ e = - v).

(** [f 0 + f 1 + ... + f (N-1)]. *)
Fixpoint sumZ (f : nat -> Z) (N : nat) : Z :=
  match N with 0 => 0 | S N' => sumZ f N' + f N' end.

(** ** A line *)

Definition frac_digits : nat := 13. (* hex digits of a binary64 fraction *)

(** [cs] is the line [[-]0x1[.F]pSX N W_0 ... W_(k l - 1)] of
    [(S0, ex, n, B)]. *)
Definition line_text (cs : list int) (S0 ex n : Z) (B : list Z) : Prop :=
  exists Sg Dot F f Xs N Ws ws,
  cs = Sg ++ [c_0; c_x; c_1] ++ Dot ++ F ++ c_p :: Xs ++
    c_space :: N ++ concat (map (fun W => c_space :: W) Ws) /\
    (* the fraction: absent, or a dot and 0 to 13 digits of value f *)
    (Dot = [] /\ F = [] \/ Dot = [c_dot]) /\
    hexs_text F f /\ (length F <= frac_digits)%nat /\
    (* S0 = +-(16^13 + f 16^(13 - |F|)), minus if the line starts by '-' *)
    (Sg = [] /\ S0 = 16 ^ Z.of_nat frac_digits +
      f * 16 ^ (Z.of_nat frac_digits - Z.of_nat (length F)) \/
      Sg = [c_minus] /\ S0 = - (16 ^ Z.of_nat frac_digits +
        f * 16 ^ (Z.of_nat frac_digits - Z.of_nat (length F)))) /\
    (* the exponent: ex = +X or -X *)
    exp_text Xs ex /\
    (* n = N *)
    dec_text N n /\
    (* the k l words w_j *)
    length Ws = (k * Z.to_nat l)%nat /\ Forall2 word_text Ws ws /\
    (* B_i = sum_(j < l) w_(i l + j) beta^j *)
    B = map (fun i => sumZ (fun j => nth (i * Z.to_nat l + j) ws 0 *
      beta ^ Z.of_nat j) (Z.to_nat l))
      (seq 0 k).

(** ** Proofs *)

Lemma decdP c d : decd c = Some d -> decdigit c d.
Proof.
  unfold decd, decdigit.
  destruct ((c_0 <=? c)%uint63 && (c <=? c_9)%uint63) eqn:E1; [|discriminate].
  intros [= <-]; apply andb_prop in E1 as [E1 E2].
  apply leb_spec in E1, E2.
  pose proof (to_Z_bounded c) as Hc.

```

```

change (to_Z c_0) with 48 in *; change (to_Z c_9) with 57 in *.
rewrite sub_spec; change (to_Z c_0) with 48.
rewrite Z.mod_small; lia.
Qed.

Lemma hexdP c d : hexd c = Some d -> hexdigit c d.
Proof.
unfold hexd, hexdigit.
destruct ((c_0 <=? c)%uint63 && (c <=? c_9)%uint63) eqn:E1.
{ intros H; left; apply decdP; unfold decd; rewrite E1; exact H. }
destruct ((c_a <=? c)%uint63 && (c <=? c_f)%uint63) eqn:E2; [|discriminate].
intros [= <-]; right; apply andb_prop in E2 as [E2 E3].
apply leb_spec in E2, E3.
pose proof (to_Z_bounded c) as Hc.
change (to_Z c_a) with 97 in *; change (to_Z c_f) with 102 in *.
rewrite sub_spec; change (to_Z c_a) with 97.
rewrite Z.mod_small; lia.
Qed.

Lemma digitsP d b acc cs v : digits d b acc cs = Some v ->
  exists ds, Forall2 (fun c x => d c = Some x) cs ds /\
    v = acc * b ^ Z.of_nat (length cs) + value b ds.
Proof.
revert acc; induction cs as [|c cs IH]; intros acc; cbn [digits].
- intros [= <-]; exists []; split; [constructor|simpl; ring].
- destruct (d c) as [x] eqn:Hc; [|discriminate].
  intros H; destruct (IH _ H) as [ds [H1 H2]].
  exists (x :: ds); split; [constructor; auto|].
  cbn [value List.length].
  rewrite <- (Forall2_length H1), H2, Nat2Z.inj_succ, Z.pow_succ_r by lia.
  ring.
Qed.

Lemma decvP cs v : decv cs = Some v -> dec_text cs v.
Proof.
unfold decv; destruct cs as [|c cs']; [discriminate|].
intros H; apply digitsP in H as [ds [H1 H2]].
split; [discriminate|]; exists ds; split.
- eapply Forall2_impl; [|exact H1]; intros ? ? ?; apply decdP; assumption.
- rewrite H2; ring.
Qed.

Lemma hexvP cs v : hexv cs = Some v -> hex_text cs v.
Proof.
unfold hexv; destruct cs as [|c cs']; [discriminate|].
intros H; apply digitsP in H as [ds [H1 H2]].
split; [discriminate|]; exists ds; split.
- eapply Forall2_impl; [|exact H1]; intros ? ? ?; apply hexdP; assumption.
- rewrite H2; ring.
Qed.

Lemma wordvP cs w : wordv cs = Some w -> word_text cs w.
Proof.
unfold wordv; destruct cs as [|c0 [|cx cs']]; try discriminate.
destruct (c0 =? c_0)%uint63 eqn:E0; [|discriminate].
destruct (cx =? c_x)%uint63 eqn:Ex; [|discriminate].

```

```

apply eqb_spec in E0, Ex; subst; intros H.
exists cs'; split; [reflexivity|apply hexvP; exact H].
Qed.

Lemma wordsvP fs ws : wordsv fs = Some ws -> Forall2 word_text fs ws.
Proof.
revert ws; induction fs as [|f fs IH]; intros ws; simpl.
- intros [= <-]; constructor.
- destruct (wordv f) as [w|] eqn:Hf; [|discriminate|.
  destruct (wordsv fs) as [ws'|] eqn:Hfs; [|discriminate|.
    intros [= <-]; constructor; [apply wordvP|apply IH]; auto.
Qed.

Lemma split_pP cs a b : split_p cs = (a, b) -> b <> [] ->
  cs = a ++ c_p :: b.
Proof.
revert a; induction cs as [|c cs IH]; intros a; simpl.
- intros [= <- <-]; contradiction.
- destruct (c =? c_p)%uint63 eqn:Ec.
  + apply eqb_spec in Ec; subst; intros [= <- <-]; reflexivity.
  + destruct (split_p cs) as [a' b'] eqn:Hs.
    intros [= <- <-] Hb; simpl; f_equal; apply IH; auto.
Qed.

Lemma expvP cs e : expv cs = Some e -> exp_text cs e.
Proof.
unfold expv; destruct cs as [|c cs']; [discriminate|].
destruct (c =? c_plus)%uint63 eqn:Ep.
- apply eqb_spec in Ep; subst; intros H.
  exists cs', e; split; [apply decvP; exact H|left; auto].
- destruct (c =? c_minus)%uint63 eqn:Em; [|discriminate|.
  apply eqb_spec in Em; subst.
  destruct (decv cs') as [v|] eqn:Hv; [|discriminate|.
    intros [= <-]; exists cs', v; split; [apply decvP; exact Hv|right; auto].
Qed.

(** The significand  $[16^{13} + f \cdot 16^{(13 - |F|)}]$  of  $[0x1.F]$ . *)
Definition signif (F : list int) (f : Z) : Z :=
  16 ^ Z.of_nat frac_digits +
  f * 16 ^ (Z.of_nat frac_digits - Z.of_nat (length F)).

Lemma fracvP cs M e : fracv cs = Some (M, e) ->
  exists Dot F f Xs, cs = Dot ++ F ++ c_p :: Xs /\
  (Dot = [] /\ F = [] \/ Dot = [c_dot]) /\
  hexs_text F f /\ (length F <= frac_digits)%nat /\
  M = signif F f /\ exp_text Xs e.
Proof.
unfold fracv; destruct (split_p cs) as [fr ex] eqn:Hs.
assert (Hex : ex <> [] -> cs = fr ++ c_p :: ex).
{ intros H; apply split_pP; [exact Hs|exact H]. }
destruct fr as [|cd fr'].
- destruct (expv ex) as [e'|] eqn:He; [|discriminate|.
  intros [= <- <-].
  exists [], [], 0, ex; split; [|split; [|split; [|split; [|split; [|split]]]]|.
  + apply Hex; intros ->; discriminate He.
  + left; auto.

```

```

+ exists []; split; [constructor|reflexivity].
+ unfold frac_digits; simpl; lia.
+ unfold signif; rewrite Z.mul_0_l, Z.add_0_r; reflexivity.
+ apply expvP; exact He.
- destruct ((cd =? c_dot)%uint63 && Nat.leb (length fr') 13) eqn:Ec;
  [|discriminate].
  apply andb_prop in Ec as [Ed Hl].
  apply eqb_spec in Ed; apply Nat.leb_le in Hl; subst cd.
  destruct (digits hexd 16 1 fr') as [f1] eqn:Hd; [|discriminate].
  destruct (expv ex) as [e'] eqn:He; [|discriminate].
  intros [= <- <-].
  apply digitsP in Hd as [ds [H1 H2]].
  exists [c_dot], fr', (value 16 ds), ex.
  split; [|split; [|split; [|split; [|split; [|split]]]]].
  + rewrite Hex; [reflexivity|intros ->; discriminate He].
  + right; reflexivity.
  + exists ds; split; [|reflexivity].
    eapply Forall2_impl; [|exact H1]; intros ? ? ?; apply hexdP; assumption.
  + exact Hl.
+ change (f1 * 16 ^ Z.of_nat (13 - length fr') =
  signif fr' (value 16 ds)).
  unfold signif; rewrite H2, (Nat2Z.inj_sub _ _ Hl); unfold frac_digits.
  set (L := Z.of_nat (length fr')).
  assert (HL : 0 <= L <= 13) by (unfold L; lia).
  replace (16 ^ Z.of_nat 13) with (16 ^ L * 16 ^ (13 - L))
    by (rewrite <- Z.pow_add_r by lia; f_equal; lia).
  change (Z.of_nat 13) with 13; ring.
+ apply expvP; exact He.
Qed.

(** [x0v] after the optional minus sign, [sg] telling if it was there. *)
Lemma x0v_body (sg : bool) (cs1 : list int) (S0 e : Z) :
  (match cs1 with
  | c0 :: cx :: c1 :: cs' =>
    if (c0 =? c_0)%uint63 && (cx =? c_x)%uint63 && (c1 =? c_1)%uint63 then
      match fracv cs' with
      | Some (M, e') => Some (if sg then - M else M, e')
      | None => None
    end
  else None
  | _ => None
  end : option (Z * Z)) = Some (S0, e) ->
  exists Dot F f Xs, cs1 = [c_0; c_x; c_1] ++ Dot ++ F ++ c_p :: Xs /\
  (Dot = [] /\ F = [] \/ Dot = [c_dot]) /\
  hexs_text F f /\ (length F <= frac_digits)%nat /\
  S0 = (if sg then - signif F f else signif F f) /\ exp_text Xs e.
Proof.
destruct cs1 as [|c0 [|cx [|c1 cs']]]; try discriminate.
destruct (c0 =? c_0)%uint63 eqn:E0; [|discriminate].
destruct (cx =? c_x)%uint63 eqn:Ex; [|discriminate].
destruct (c1 =? c_1)%uint63 eqn:E1; [|discriminate]; cbn [andb].
apply eqb_spec in E0, Ex, E1; subst.
destruct (fracv cs') as [[M e']] eqn:Hf; [|discriminate].
intros [= <- <-].
apply fracvP in Hf as (Dot & F & f & Xs & -> & HD & HF & HL & -> & HX).
exists Dot, F, f, Xs; repeat split; auto.

```

Qed.

```
Lemma x0vP cs S0 e : x0v cs = Some (S0, e) ->
  exists Sg Dot F f Xs,
  cs = Sg ++ [c_0; c_x; c_1] ++ Dot ++ F ++ c_p :: Xs /\
  (Dot = [] /\ F = [] \/ Dot = [c_dot]) /\
  hexs_text F f /\ (length F <= frac_digits)%nat /\
  (Sg = [] /\ S0 = signif F f \/ Sg = [c_minus] /\ S0 = - signif F f) /\
  exp_text Xs e.
```

Proof.

```
unfold x0v; destruct cs as [|c cs0]; [discriminate|].
destruct (c =? c_minus)%uint63 eqn:Em.
- intros H; apply (x0v_body true cs0) in H.
  destruct H as (Dot & F & f & Xs & -> & HD & HF & HL & HS & HX).
  apply eqb_spec in Em; subst.
  exists [c_minus], Dot, F, f, Xs; repeat split; auto.
- intros H; apply (x0v_body false (c :: cs0)) in H.
  destruct H as (Dot & F & f & Xs & Hc & HD & HF & HL & HS & HX).
  exists [], Dot, F, f, Xs; repeat split; auto.
```

Qed.

```
Lemma fields_nil cur cs : fields cur cs <> [].
```

Proof.

```
revert cur; induction cs as [|c cs IH]; intros cur; simpl; [discriminate|].
destruct (c =? c_space)%uint63; [discriminate|apply IH].
Qed.
```

```
Lemma fieldsP cur cs f fs : fields cur cs = f :: fs ->
```

```
  rev cur ++ cs = f ++ concat (map (fun W => c_space :: W) fs).
```

Proof.

```
revert cur f fs; induction cs as [|c cs IH]; intros cur f fs; simpl.
- intros [= <- <-]; simpl; rewrite !app_nil_r; reflexivity.
- destruct (c =? c_space)%uint63 eqn:Ec.
  + apply eqb_spec in Ec; subst; intros [= <- <-].
    destruct (fields [] cs) as [|g gs] eqn:Hg; [now apply fields_nil in Hg|].
    apply IH in Hg; simpl in Hg; rewrite Hg; reflexivity.
  + intros H; apply IH in H; rewrite <- H; simpl.
    rewrite <- app_assoc; reflexivity.
```

Qed.

```
Lemma sumZ_ext f g N : (forall j, f j = g j) -> sumZ f N = sumZ g N.
```

Proof. intros H; induction N; simpl; [reflexivity|rewrite IHN, H; ring]. Qed.

```
Lemma sumZ_shift f N : sumZ f (S N) = f 0 + sumZ (fun j => f (S j)) N.
```

Proof.

```
induction N; [simpl; ring|].
change (sumZ f (S (S N))) with (sumZ f (S N) + f (S N)).
rewrite IHN; simpl; ring.
```

Qed.

```
Lemma sumZ_scale c f N :
```

```
  sumZ (fun j => c * f j) N = c * sumZ f N.
```

Proof. induction N; simpl; [ring|rewrite IHN; ring]. Qed.

```
Lemma wsum_firstn n ws : wsum (firstn n ws) =
```

```
  sumZ (fun j => nth j ws 0 * beta ^ Z.of_nat j) n.
```

```

Proof.
revert ws; induction n as [|n IH]; intros ws; [reflexivity|].
rewrite sumZ_shift; destruct ws as [|w ws].
- rewrite (sumZ_ext _ (fun j => 0 * beta ^ Z.of_nat (S j)))
  by (intros [|j]; reflexivity).
  rewrite sumZ_scale; unfold wsum; cbn [firstn nth fold_right]; ring.
- change (wsum (firstn (S n) (w :: ws))) with
  (w + beta * wsum (firstn n ws)).
  rewrite IH.
  rewrite (sumZ_ext (fun j => nth (S j) (w :: ws) 0 * beta ^ Z.of_nat (S j))
    (fun j => beta * (nth j ws 0 * beta ^ Z.of_nat j))).
  + rewrite (sumZ_scale beta (fun j => nth j ws 0 * beta ^ Z.of_nat j)).
    cbn [nth]; change (Z.of_nat 0) with 0; ring.
  + intros j; cbn [nth]; rewrite Nat2Z.inj_succ, Z.pow_succ_r by lia; ring.
Qed.

```

```

Lemma groupP c ws : group c ws =
  map (fun i => sumZ (fun j => nth (i * Z.to_nat l + j) ws 0 *
    beta ^ Z.of_nat j) (Z.to_nat l))
    (seq 0 c).

```

```

Proof.
revert ws; induction c as [|c IH]; intros ws; [reflexivity|].
change (group (S c) ws) with
  (wsum (firstn (Z.to_nat l) ws) :: group c (skipn (Z.to_nat l) ws)).
rewrite IH, wsum_firstn; cbn [seq map]; rewrite <- seq_shift, map_map.
f_equal; apply map_ext; intros i; apply sumZ_ext; intros j.
rewrite nth_skipn; f_equal; f_equal; lia.
Qed.

```

(** ** The theorem *)

```

(** Whatever [parse] returns is what the line says. *)
Theorem parse_sound s S0 ex n B :
  parse s = Some (S0, ex, n, B) -> line_text (to_list s) S0 ex n B.

```

```

Proof.
unfold parse.
destruct (fields [] (to_list s)) as [|fx [|fn fw]] eqn:Hf; try discriminate.
destruct (Nat.eqb_spec (length fw) (k * Z.to_nat l)) as [Hl|];
  [|discriminate|.
destruct (x0v fx) as [[S e]] eqn:Hx; [|discriminate|.
destruct (decv fn) as [n'|] eqn:Hn; [|discriminate|.
destruct (wordsv fw) as [ws|] eqn:Hw; [|discriminate|.
intros [= <- <- <- <-].
apply fieldsP in Hf; simpl in Hf.
apply x0vP in Hx as (Sg & Dot & F & f & Xs & -> & HD & HF & HL & HS & HX).
exists Sg, Dot, F, f, Xs, fn, fw, ws.
split; [|split; [|split; [|split; [|split; [|split; [|split; [|split; [|split;
  [|split]]]]]]]]]; auto.
- rewrite Hf, <- !app_assoc; reflexivity.
- apply decvP; exact Hn.
- apply wordsvP; exact Hw.
- exact (groupP k ws).
Qed.

```

(** ** A short example *)

```

(** Six words [0x0], an [l]-word group of zeros. *)
Definition zeros6 : string := " 0x0 0x0 0x0 0x0 0x0 0x0"%pstring.

(** Eight groups of zeros. *)
Definition zeros48 : string :=
  cat zeros6 (cat zeros6 (cat zeros6 (cat zeros6
    (cat zeros6 (cat zeros6 (cat zeros6 zeros6)))))).

(** [x0 = -0x1.8p-3 = -3 2^51 2^(-3 - 52)], [n = 3], words [1, 2, 0, ...]. *)
Definition example_line : string :=
  cat "-0x1.8p-3 3 0x1 0x2 0x0 0x0 0x0 0x0"%pstring zeros48.

Example parse_example : parse example_line =
  Some (- 3 * 2 ^ 51, -3, 3, [1 + 2 * beta; 0; 0; 0; 0; 0; 0; 0; 0]).
Proof. vm_compute; reflexivity. Qed.

(** [x0 = 0x1p+0 = 1] without a fraction: [S0 = 16^13], [ex = 0]. *)
Example parse_example2 :
  parse (cat "0x1p+0 7 0x0 0x0 0x0 0x0 0x0 0x0"%pstring zeros48) =
  Some (16 ^ 13, 0, 7, [0; 0; 0; 0; 0; 0; 0; 0; 0]).
Proof. vm_compute; reflexivity. Qed.

```

E ExpArith.v

```

(** * The integer tests of the checker, read in the reals *)

From Stdlib Require Import ZArith Reals Lia Lra List.
From Flocq Require Import Core.
From ExpTable9 Require Import ExpCheck.

Open Scope R_scope.

(** [flr] is the floor it is named after. *)
Lemma flrE p g fk h : (0 < fk)%Z -> (0 <= h)%Z ->
  flr p g fk h = Zfloor (IZR p * bp g / IZR fk + IZR h / 2).
Proof.
  intros Hfk Hh; unfold flr, bp.
  assert (Hfk' : IZR fk <> 0) by (apply not_0_IZR; lia).
  destruct (Z.leb_spec 0 g) as [Hg | Hg].
  - rewrite <- Zfloor_div by lia; f_equal.
    rewrite <- (IZR_Zpower radix2 g) by lia; simpl radix_val.
    rewrite !mult_IZR, plus_IZR, !mult_IZR; field; auto.
  - rewrite <- Zfloor_div.
    2: { apply Z.neq_mul_0; split; [lia]; apply Z.pow_nonzero; lia. }
    f_equal.
    (* a negative power of two is the inverse of an integer *)
    assert (Hq : bpow radix2 g = / IZR (2 ^ (- g))).
    { rewrite (IZR_Zpower radix2) by lia; rewrite bpow_opp, Rinv_inv; easy. }
    assert (H2 : IZR (2 ^ (- g)) <> 0).
    { apply not_0_IZR; apply Z.pow_nonzero; lia. }
    rewrite Hq, !mult_IZR, plus_IZR, !mult_IZR; field; auto.
Qed.

Lemma lt2_ok p f g : (1 <= p)%Z -> lt2 p f g = true -> IZR p * bp f < bp g.
Proof.

```

```

intros Hp; unfold lt2, bp.
destruct (Z.leb_spec g f) as [Hgf | Hgf]; [discriminate|].
intros Hlt; apply Z.ltb_lt in Hlt.
replace (bpow radix2 g) with (bpow radix2 ((g - f) + f)) by (f_equal; ring).
rewrite bpow_plus; apply Rmult_lt_compat_r; [apply bpow_gt_0|].
rewrite <- (IZR_Zpower radix2) by lia; apply IZR_lt; exact Hlt.
Qed.

```

Lemma log2_le p f : (0 < p)%Z -> bp (Z.log2 p + f) <= IZR p * bp f.
Proof.

```

intros Hp; unfold bp; rewrite bpow_plus.
apply Rmult_le_compat_r; [apply Rlt_le, bpow_gt_0|].
rewrite <- (IZR_Zpower radix2) by (apply Z.log2_nonneg).
apply IZR_le; destruct (Z.log2_spec p Hp) as [H _]; exact H.
Qed.

```

(** [beta^l] is the power of two of exponent [lbits]. *)
Lemma beta_lE : beta_l = (2 ^ lbits)%Z.
Proof. vm_compute; reflexivity. Qed.

Lemma lbits_ge0 : (0 <= lbits)%Z.
Proof. vm_compute; discriminate. Qed.

Lemma beta_lR : IZR beta_l = bp lbits.
Proof.
rewrite beta_lE; unfold bp; rewrite (IZR_Zpower radix2); [easy|].
exact lbits_ge0.
Qed.

(** A power of a power of two. *)
Lemma bp_pow a i : bp a ^ i = bp (a * Z.of_nat i).
Proof.
induction i as [|i IH]; [now rewrite Z.mul_0_r|].
rewrite Nat2Z.inj_succ, <- Z.add_1_r, Z.mul_add_distr_l, Z.mul_1_r.
unfold bp in *; rewrite bpow_plus, <- IH; simpl; ring.
Qed.

(** Moving the start of an integer sum. *)
Lemma fold_addE (s : list Z) a :
 fold_right Z.add a s = (fold_right Z.add 0 s + a)%Z.
Proof. induction s as [|x s IH]; simpl; lia. Qed.

(** [sumR] of powers of an integer, in integers. *)
Lemma sumR_powE n N : sumR (fun i => IZR n ^ i) N =
 IZR (fold_right Z.add 0 (map (fun i => n ^ Z.of_nat i) (seq 0 N)))%Z.
Proof.
induction N as [|N IH]; [easy|].
rewrite seq_S, map_app, fold_right_app; simpl.
rewrite fold_addE, plus_IZR, <- IH, Z.add_0_r, pow_IZR; easy.
Qed.

(** A power of two, as an integer. *)
Lemma bp_IZR g : (0 <= g)%Z -> bp g = IZR (2 ^ g).
Proof.
intros Hg; unfold bp; rewrite <- (IZR_Zpower radix2) by exact Hg; easy.
Qed.

```

(** A negative power of two, as the inverse of an integer. *)
Lemma bp_IZRN g : (g < 0)%Z -> bp g = / IZR (2 ^ (- g)).
Proof.
intros Hg; unfold bp; rewrite (IZR_Zpower radix2) by lia.
rewrite bpow_opp, Rinv_inv; easy.
Qed.

(** [window_ok] in the reals, before the bound on [exp x1]. *)
Lemma window_okR mU fU ve ue n : window_ok mU fU ve ue n = true ->
  IZR (2 ^ (lbits - m) + sumn n) + IZR (mU * n ^ Z.of_nat k) *
  bp (fU + lbits + ue * Z.of_nat k - ve) / IZR (Z.of_nat (fact k))
  <= IZR E.
Proof.
unfold window_ok; cbv zeta.
set (X := (mU * n ^ Z.of_nat k)%Z).
set (g := (fU + lbits + ue * Z.of_nat k - ve)%Z).
set (C := (2 ^ (lbits - m) + sumn n)%Z).
set (kf := Z.of_nat (fact k)).
assert (Hkf : (0 < kf)%Z) by (pose proof (lt_0_fact k); lia).
assert (HkfR : 0 < IZR kf) by (apply IZR_lt; exact Hkf).
clearbody X g C kf; generalize E; intros EE.
destruct (Z.leb_spec 0 g) as [Hg | Hg]; intros H; apply Z.leb_le, IZR_le in H;
  rewrite ?plus_IZR, ?mult_IZR, ?plus_IZR, ?mult_IZR in H.
- rewrite bp_IZR by exact Hg.
  (* field and lra only see real variables, or the kernel check is slow *)
  generalize (IZR X) (IZR C) (IZR (2 ^ g)) (IZR EE) H.
  generalize (IZR kf) HkfR; clear; intros kr Hk xr cr pr er H.
  apply Rmult_le_reg_r with (1 := Hk).
  replace ((cr + xr * pr / kr) * kr) with (xr * pr + cr * kr) by (field; lra).
  exact H.
- rewrite bp_IZRN by exact Hg.
  assert (HP : 0 < IZR (2 ^ (- g))).
  { apply IZR_lt; apply Z.pow_pos_nonneg; lia. }
  generalize (IZR X) (IZR C) (IZR (2 ^ (- g))) (IZR EE) H HP.
  generalize (IZR kf) HkfR; clear; intros kr Hk xr cr pr er H Hp.
  apply Rmult_le_reg_r with (r := kr * pr); [now apply Rmult_lt_0_compat|].
  replace ((cr + xr * / pr / kr) * (kr * pr))
    with (xr + cr * kr * pr) by (field; lra).
  lra.
Qed.

(** (H_E), from an upper bound [mU 2^fU] of [y = exp x1]. *)
Lemma window_okP mU fU ve ue n y :
  (m <= lbits)%Z -> (0 <= n)%Z -> 0 <= y <= IZR mU * bp fU ->
  window_ok mU fU ve ue n = true ->
  IZR beta_l * (bp (- m) + y * (IZR n * bp ue) ^ k / (INR (fact k) * bp ve))
  + sumR (fun i => IZR n ^ i) k <= IZR E.
Proof.
intros Hm Hn Hy Hw.
eapply Rle_trans; [| exact (window_okR _ _ _ _ Hw)].
rewrite sumR_powE; fold (sumn n).
rewrite plus_IZR, mult_IZR, <- pow_IZR, <- INR_IZR_INZ.
rewrite <- bp_IZR by lia.
rewrite beta_lR.
rewrite Rpow_mult_distr, bp_pow.

```

```

replace (bp (lbits - m)) with (bp lbits * bp (- m))
  by (unfold bp; rewrite <- bpow_plus; f_equal; ring).
replace (bp (fU + lbits + ue * Z.of_nat k - ve))
  with (bp fU * bp lbits * bp (ue * Z.of_nat k) / bp ve)
  by (unfold bp, Rdiv; rewrite <- bpow_opp, <- !bpow_plus; f_equal; ring).
assert (HB : 0 < bp lbits) by apply bpow_gt_0.
assert (HU : 0 < bp (ue * Z.of_nat k)) by apply bpow_gt_0.
assert (HV : 0 < bp ve) by apply bpow_gt_0.
assert (HK : 0 < INR (fact k)) by apply INR_fact_lt_0.
assert (HN : 0 <= IZR n ^ k) by (apply pow_le, IZR_le; exact Hn).
(* only real variables are left *)
revert Hy HB HU HV HK HN.
generalize (bp lbits) (bp (- m)) (bp (ue * Z.of_nat k)) (bp ve).
generalize (INR (fact k)) (IZR n ^ k) (IZR (sumn n)) (IZR mU) (bp fU).
clear; intros K N S F1 F2 B M U V Hy HB HU HV HK HN.
assert (Hc : 0 <= N * U * B / (K * V)).
{ unfold Rdiv; apply Rmult_le_pos.
  - apply Rmult_le_pos; [apply Rmult_le_pos]; lra.
  - apply Rlt_le, Rinv_0_lt_compat, Rmult_lt_0_compat; easy. }
replace (B * (M + y * (N * U) / (K * V)) + S)
  with (B * M + S + y * (N * U * B / (K * V))) by (field; lra).
replace (B * M + S + F1 * N * (F2 * B * U / V) / K)
  with (B * M + S + (F1 * F2) * (N * U * B / (K * V))) by (field; lra).
apply Rplus_le_compat_l, Rmult_le_compat_r; [exact Hc | apply Hy].
Qed.

(** A floor that is the same at both ends is the same in between. *)
Lemma Zfloor_mid a b d : a <= b <= d -> Zfloor a = Zfloor d ->
  Zfloor b = Zfloor a.
Proof.
intros [Hab Hbd] Had.
apply Zfloor_le in Hab; apply Zfloor_le in Hbd; lia.
Qed.

(** (H_A) for one coefficient, from an enclosure of [y = exp x0]. *)
Lemma coefA_ok mL fL mU fU ve ue i fk Ai y :
  fk = Z.of_nat (fact i) ->
  IZR mL * bp fL <= y <= IZR mU * bp fU ->
  coefA mL fL mU fU ve ue i fk = Some Ai ->
  (0 <= Ai < beta_l)%Z /\
  Rabs (IZR Ai - IZR beta_l *
    fracR (y * bp ue ^ i / INR (fact i) / bp ve)) < 1.
Proof.
intros Hfk Hy H.
assert (Hfk0 : (0 < fk)%Z) by (pose proof (lt_0_fact i); lia).
unfold coefA in H; cbv zeta in H.
set (gi := (lbits + ue * Z.of_nat i - ve)%Z) in H.
set (r := flr mL (fL + gi) fk 1) in H.
set (q := flr mL (fL + gi - lbits) fk 0) in H.
destruct ( _ && _ ) eqn:Hb; [|discriminate].
injection H as <-.
rewrite !Bool.andb_true_iff, Z.eqb_eq, Z.eqb_eq, Z.leb_le, Z.ltb_lt in Hb.
destruct Hb as [[H1 H2] H3] H4].
split; [lia].
unfold r, q in *.
rewrite !flrE in H1, H2 by lia.

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rewrite !flrE by lia.
clear H3 H4.
assert (HfkR : 0 < IZR fk) by (apply IZR_lt; exact Hfk0).
assert (HB : 0 < bp lbits) by apply bpow_gt_0.
(* the argument of fracR is  $y (2^{gi} / i!) / \beta^l$  *)
assert (Hc :  $y * bp \text{ ue } ^i / \text{INR (fact i)} / bp \text{ ve } =$ 
         $y * (bp \text{ gi } / \text{IZR fk}) / bp \text{ lbits}$ ).
{ rewrite INR_IZR_INZ, <- Hfk; rewrite bp_pow.
  unfold gi; replace (lbits + ue * Z.of_nat i - ve)%Z
    with (lbits + ue * Z.of_nat i + - ve)%Z by ring.
  assert (HV : 0 < bp ve) by apply bpow_gt_0.
  unfold bp in *; rewrite !bpow_plus, bpow_opp; field; lra. }
rewrite Hc; clear Hc; clearbody gi.
assert (Ha : forall z f, IZR z * bp (f + gi) / IZR fk =
        IZR z * bp f * (bp gi / IZR fk)).
{ intros z f; unfold bp; rewrite bpow_plus; field; lra. }
assert (Hb : forall z f, IZR z * bp (f + gi - lbits) / IZR fk + 0 / 2 =
        IZR z * bp f * (bp gi / IZR fk) / bp lbits).
{ intros z f; replace (f + gi - lbits)%Z with (f + gi + - lbits)%Z by ring.
  unfold bp; rewrite !bpow_plus, bpow_opp.
  unfold bp in HB; field; lra. }
rewrite !Ha, !Hb in *; clear Ha Hb.
assert (Hc : 0 < bp gi / IZR fk).
{ apply Rdiv_lt_0_compat; [apply bpow_gt_0 | exact HfkR]. }
rewrite minus_IZR, mult_IZR, beta_lR.
unfold fracR.
(* only real variables are left *)
revert Hy H1 H2 HB Hc.
generalize (IZR mL * bp fL) (IZR mU * bp fU) (bp gi / IZR fk) (bp lbits).
clear; intros lo hi c B Hy H1 H2 HB Hc.
assert (Hlo : lo * c <= y * c) by (apply Rmult_le_compat_r; lra).
assert (Hhi : y * c <= hi * c) by (apply Rmult_le_compat_r; lra).
rewrite <- (Zfloor_mid (lo * c / B) (y * c / B) (hi * c / B));
  [| | exact H2].
2: { unfold Rdiv; split; apply Rmult_le_compat_r; try lra;
    apply Rlt_le, Rinv_0_lt_compat; exact HB. }
rewrite <- (Zfloor_mid (lo * c + 1 / 2) (y * c + 1 / 2) (hi * c + 1 / 2));
  [| lra | exact H1].
pose proof (Zfloor_lb (y * c + 1 / 2)).
pose proof (Zfloor_ub (y * c + 1 / 2)).
replace (B * (y * c / B - IZR (Zfloor (y * c / B))))
  with (y * c - IZR (Zfloor (y * c / B)) * B) by (field; lra).
apply Rabs_def1; lra.
Qed.

(** (H_A) for the [c] coefficients from the [i]-th on. *)
Lemma coefs_ok mL fL mU fU ve ue y c i fk As :
  fk = Z.of_nat (fact i) ->
  IZR mL * bp fL <= y <= IZR mU * bp fU ->
  coefs mL fL mU fU ve ue i fk c = Some As ->
  length As = c /\
  forall j, (j < c)%nat ->
    (0 <= nth j As 0%Z < beta_l)%Z /\
    Rabs (IZR (nth j As 0%Z) - IZR beta_l *
      fracR (y * bp ue ^ (i + j) / INR (fact (i + j)) / bp ve)) < 1.
Proof.

```

```

intros Hfk Hy; revert i fk As Hfk.
induction c as [|c IH]; intros i fk As Hfk H; cbn [coefs] in H.
- injection H as <-; split; [easy | intros j Hj; lia].
- destruct (coefA mL fL mU fU ve ue i fk) as [Ai] eqn:HA; [|discriminate].
  destruct (coefs mL fL mU fU ve ue (S i) (fk * Z.of_nat (S i)) c) as [As']
    eqn:HAs; [|discriminate].
  injection H as <-.
  (* the next factorial *)
  assert (Hfk' : (fk * Z.of_nat (S i))%Z = Z.of_nat (fact (S i))).
  { change (fact (S i)) with (S i * fact i)%nat.
    rewrite Nat2Z.inj_mul, Hfk; ring. }
  destruct (IH _ _ _ Hfk' HAs) as [Hl HAs'].
  split; [simpl; lia].
  intros [|j] Hj.
  + rewrite Nat.add_0_r; exact (coefA_ok _ _ _ _ _ _ _ _ _ Hfk Hy HA).
  + replace (i + S j)%nat with (S i + j)%nat by lia.
    apply HAs'; lia.
Qed.

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F ExpEnc1.v

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(** * The enclosure of exp given by Interval *)

From Stdlib Require Import ZArith Reals Lia Lra.
From Flocq Require Import Core.
From Interval Require Import Xreal Basic Specific_bigint Specific_ops.
From ExpTable9 Require Import ExpCheck.

Open Scope R_scope.

(* A float with a radix-2 exponent is its signed significand times 2^e. *)
Lemma FtoR_sgn s p e :
  FtoR radix2 s p e = IZR (if s then Z.neg p else Z.pos p) * bp e.
Proof.
unfold bp; destruct s, e as [|q|q]; cbv beta iota zeta delta [FtoR bpow].
all: try rewrite mult_IZR; try lra; reflexivity.
Qed.

(* A positive float with a radix-2 exponent is p 2^e. *)
Lemma FtoR_pos p e : FtoR radix2 false p e = IZR (Z.pos p) * bp e.
Proof. exact (FtoR_sgn false p e). Qed.

(* The value of a float read by getS. *)
Lemma getS_val f p e : getS f = Some (p, e) ->
  SFBI2.toX f = Xreal (IZR p * bp e).
Proof.
unfold getS, SFBI2.toX.
destruct (SFBI2.toF f) as [| |s p' e']; try discriminate.
intros H; injection H; intros <- <-; simpl FtoX; rewrite <- FtoR_sgn.
reflexivity.
Qed.

(* The value of a float read by getF. *)
Lemma getF_val f p e : getF f = Some (p, e) ->
  SFBI2.toX f = Xreal (IZR (Z.pos p) * bp e).

```

```

Proof.
unfold getF, SFBI2.toX.
destruct (SFBI2.toF f) as [| [|] p' e']; try discriminate.
intros H; injection H; intros <- <-; simpl FtoX; rewrite <- FtoR_pos.
reflexivity.
Qed.

Lemma encl_ok N ue mL fL mU fU : encl N ue = Some (mL, fL, mU, fU) ->
  (0 < mL)%Z /\ (0 < mU)%Z /\
  IZR mL * bp fL <= exp (IZR N * bp ue) <= IZR mU * bp fU.
Proof.
unfold encl.
destruct (SFBI2.valid_lb (xpt N ue)) eqn:VL; [|discriminate].
destruct (SFBI2.valid_ub (xpt N ue)) eqn:VU; [|discriminate].
destruct (getS (xpt N ue)) as [|p e]| eqn:GX; [|discriminate].
destruct (Z.eqb_spec p N) as [HN|]; [|discriminate].
destruct (Z.eqb_spec e ue) as [He|]; [|discriminate].
simpl andb; cbv iota.
(* The point interval contains N 2^ue. *)
assert (Hc : Interval.contains
  (I.convert (Float.Ibnd (xpt N ue) (xpt N ue))) (Xreal (IZR N * bp ue))).
{ unfold I.convert; change I.F.valid_lb with SFBI2.valid_lb.
  change I.F.valid_ub with SFBI2.valid_ub; rewrite VL, VU.
  change I.F.toX with SFBI2.toX; rewrite (getS_val _ _ _ GX), HN, He.
  simpl; lra. }
generalize (I.exp_correct iprec _ _ Hc).
destruct (I.exp iprec (Float.Ibnd (xpt N ue) (xpt N ue))) as [|lo up];
  [discriminate|].
unfold I.convert.
change I.F.valid_lb with SFBI2.valid_lb.
change I.F.valid_ub with SFBI2.valid_ub.
change (SFBI2.valid_lb lo) with true; rewrite Bool.andb_true_l.
destruct (SFBI2.valid_ub up); [|intros _ H; discriminate H].
destruct (getF lo) as [|pL fL'|] eqn:GL; [|intros _ H; discriminate H].
destruct (getF up) as [|pU fU'|] eqn:GU; [|intros _ H; discriminate H].
change I.F.toX with SFBI2.toX.
rewrite (getF_val _ _ _ GL), (getF_val _ _ _ GU).
intros [H1 H2] H; injection H; intros <- <- <- <-.
split; [lia|split; [lia|]].
split; assumption.
Qed.

```

G ExpTaylor.v

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(** * The Taylor remainder of exp: condition 4, (H_T) *)

From Stdlib Require Import ZArith Reals Lia Lra.
From Coquelicot Require Import Coquelicot.
From ExpTable9 Require Import ExpCheck.

Open Scope R_scope.

(* Every derivative of exp is exp. *)
Lemma Derive_n_exp (m : nat) (t : R) : Derive_n exp m t = exp t.
Proof.

```

```

revert t; induction m as [|m IH]; intros t; [reflexivity|].
simpl; rewrite (Derive_ext _ _ IH).
now apply is_derive_unique, is_derive_exp.
Qed.

(* exp is differentiable [m] times everywhere. *)
Lemma ex_derive_n_exp (m : nat) (t : R) : ex_derive_n exp m t.
Proof.
destruct m as [|m]; [exact I|simpl].
apply (ex_derive_ext exp); [intros s; now rewrite Derive_n_exp|].
eexists; apply is_derive_exp.
Qed.

(* [sumR] with [N + 1] terms is the Stdlib [sum_f_R0] up to [N]. *)
Lemma sumR_sum_f (f : nat -> R) (N : nat) : sumR f (S N) = sum_f_R0 f N.
Proof.
induction N as [|N IH]; [simpl; ring|].
change (sumR f (S N) + f (S N) = sum_f_R0 f N + f (S N)); now rewrite IH.
Qed.

(** For [h >= 0], the Taylor polynomial of degree [N-1] of [exp] at [x0]
    is below [exp (x0 + h)] by at most [exp (x0 + h) h^N / N!]. *)
Lemma taylor_exp (x0 h : R) (N : nat) : 0 <= h ->
  Rabs (exp (x0 + h) - sumR (fun i => exp x0 * h ^ i / INR (fact i)) N)
  <= exp (x0 + h) * h ^ N / INR (fact N).
Proof.
intros Hh.
destruct N as [|N].
  simpl; rewrite Rminus_0_r, Rabs_right by (left; apply exp_pos); lra.
destruct Hh as [Hh|<-].
2:{ (* At [h = 0] the sum is [exp x0] and the bound is [0]. *)
  rewrite Rplus_0_r, sumR_sum_f, pow_i by lia.
  replace (sum_f_R0 _ N) with (exp x0).
  rewrite Rminus_diag, Rabs_R0, Rmult_0_r; unfold Rdiv; lra.
  induction N as [|N IH]; [simpl; field|].
  rewrite tech5, <- IH, pow_i by lia; unfold Rdiv; ring. }
(* Lagrange: the remainder is [h^(N+1) / (N+1)! exp z], [z] in the range. *)
destruct (Taylor_Lagrange exp N x0 (x0 + h)) as [z [Hz Hz']]; [lra|].
  intros t _ m _; apply ex_derive_n_exp.
rewrite sumR_sum_f.
replace (sum_f_R0 _ N) with
  (sum_f_R0 (fun m => (x0 + h - x0) ^ m / INR (fact m) * Derive_n exp m x0)
    N).
2:{ apply sum_eq; intros i _; rewrite Derive_n_exp.
  replace (x0 + h - x0) with h by ring; unfold Rdiv; ring. }
rewrite Hz' at 1; rewrite Derive_n_exp.
replace (x0 + h - x0) with h by ring.
assert (0 < INR (fact (S N))) by apply INR_fact_lt_0.
assert (0 < h ^ S N) by (apply pow_lt; lra).
rewrite Rplus_minus_l, Rabs_right.
2:{ apply Rle_ge; unfold Rdiv; apply Rmult_le_pos; [|left; apply exp_pos|.
  apply Rmult_le_pos; [lra|left; apply Rinv_0_lt_compat; lra]. }
rewrite (Rmult_comm (exp _)); unfold Rdiv.
assert (exp z < exp (x0 + h)) by (apply exp_increasing; lra).
assert (0 < / INR (fact (S N))) by (apply Rinv_0_lt_compat; lra).
assert (0 < h ^ S N * / INR (fact (S N))) by (apply Rmult_lt_0_compat; lra).

```

```

nra.
Qed.

(* Two sums with equal terms are equal. *)
Lemma sumR_ext (f g : nat -> R) (N : nat) :
  (forall i, f i = g i) -> sumR f N = sumR g N.
Proof.
intros Hfg; induction N as [|N IH]; simpl; [easy|now rewrite IH, Hfg].
Qed.

(* A common divisor comes out of the sum. *)
Lemma sumR_div (f : nat -> R) (v : R) (N : nat) :
  sumR (fun i => f i / v) N = sumR f N / v.
Proof.
induction N as [|N IH]; simpl; [|rewrite IH]; unfold Rdiv; ring.
Qed.

(** Condition 4 of [line_ok], for [j] in [[0, n]]. *)
Lemma HT_ok (x0 u v : R) (n j : Z) : 0 < u -> 0 < v -> (0 <= j <= n)%Z ->
  Rabs (exp (x0 + IZR j * u) / v - sumR (fun i => a x0 u i / v * IZR j ^ i) k)
  <= exp (x0 + IZR n * u) * (IZR n * u) ^ k / (INR (fact k) * v).
Proof.
intros Hu Hv [Hj Hjn].
apply IZR_le in Hj; apply IZR_le in Hjn.
assert (Hh : 0 <= IZR j * u) by (apply Rmult_le_pos; lra).
(* The sum is the Taylor sum at [h = j u], divided by [v]. *)
rewrite (sumR_ext _ (fun i => exp x0 * (IZR j * u) ^ i / INR (fact i) / v)).
2:{ intros i; unfold a; rewrite Rpow_mult_distr; field.
  split; [apply not_0_INR, fact_neq_0|lra]. }
rewrite (sumR_div (fun i => exp x0 * (IZR j * u) ^ i / INR (fact i))).
set (Sj := sumR _ k).
replace (exp (x0 + IZR j * u) / v - Sj / v) with
  ((exp (x0 + IZR j * u) - Sj) / v) by (field; lra).
rewrite Rabs_div, (Rabs_right v) by lra.
assert (0 < INR (fact k)) by apply INR_fact_lt_0.
replace (_ / (INR (fact k) * v)) with
  (exp (x0 + IZR n * u) * (IZR n * u) ^ k / INR (fact k) / v)
  by (field; lra).
apply Rmult_le_compat_r; [left; apply Rinv_0_lt_compat; lra|].
eapply Rle_trans; [apply taylor_exp; lra|].
(* The bound grows with [j], so [j = n] is the worst case. *)
unfold Rdiv; apply Rmult_le_compat_r; [left; apply Rinv_0_lt_compat; lra|].
apply Rmult_le_compat.
- left; apply exp_pos.
- now apply pow_le.
- destruct (Rle_lt_or_eq_dec _ _ Hjn) as [Hlt|Heq]; [|rewrite Heq; lra].
  left; apply exp_increasing; nra.
- apply pow_incr; split; [lra|]; nra.
Qed.

```